

Model details

Data Harmonization

The empirical analysis is based on a longitudinal dataset covering the period from 2007 to 2018. A primary challenge in integrating the annual socio-economic indicators (livestock dynamics and sugarcane production) with the environmental indicators (mangrove and wetland cover) lies in their asymmetric temporal resolutions. While the agricultural data are recorded annually, ecological monitoring yields sparse observations; mangrove cover is systematically recorded at five-year intervals, whereas wetland data exhibit irregular temporal gaps with observations restricted to specific years (2007, 2011, 2016, and 2018).

To construct a balanced annual time series without discarding the high-frequency variation of the economic variables, deterministic time-series imputation algorithms were implemented using the `imputeTS` package in R. A uniform piecewise linear interpolation strategy was applied to both environmental variables:

This approach assumes a constant, incremental rate of change between the available observation nodes. For mangrove dynamics, linear interpolation preserves the long-term deterministic time-trend typical of slow-moving successional ecosystems while avoiding arbitrary polynomial fluctuations. Similarly, for the irregularly spaced wetland observations, a piecewise linear approximation was selected as a conservative estimation method that restricts the imputed values strictly within the bounds of the empirical data points, thereby preventing unrealistic statistical overshoots.

Finally, because interpolation is mathematically impossible without a forward anchor point, the time series was truncated at the year 2018. Restricting the final analytical sample to the 2007–2018 window ensures that all environmental metrics are strictly derived from observed interpolation intervals, thereby eliminating the need for forecasting or carrying forward observations into the 2019–2020 period. The finalized, balanced dataset was exported to an RDS format for subsequent econometric modeling.

Premise A: land use effect on production

The objective was to evaluate how land use affects fishery production while accounting for structural dependencies (species/locations via `nombre_principal` and time via `year`).

1. Baseline Approach (m1): A naive OLS linear model ignored grouping entirely, resulting in the worst fit (highest AIC/BIC).
2. Handling Grouping & Singularity (m2, m3, m4): Introducing random intercepts for both species and year (m2, m3) triggered singularity issues, indicating that the model was overparameterized or variance components were near zero. Removing the random effect for time (m4) resolved the singularity and improved the fit.
3. Accounting for Macro-Shocks (m5, m6): Instead of a random effect for time, a specific 2008 economic/environmental shock was introduced.

- o m5 modeled this as a uniform shock across all species.
- o m6 modeled an asymmetric macro-shock using a nested interaction (shock_2008:nombre_principal), allowing the 2008 shock to impact each species differently.

Results Comparison

Both AIC and BIC strongly favor m6, as it drastically reduces the information criteria scores despite adding extra parameters (\$df = 10\$).

Model	Degrees of Freedom (df)	AIC	BIC	Notes / Model Description
m1	4	147.02	155.40	Naive OLS baseline (No random effects)
m2	6	134.18	146.75	Crossed RE (nombre_principal + year) \$\\rightarrow\$ <i>Singular fit</i>
m3	6	134.18	146.75	Centered predictors \$\\rightarrow\$ <i>Singular fit</i>
m4	5	132.18	142.65	RE for nombre_principal only (No time effect)
m5	6	126.81	139.38	Uniform 2008 macro-shock + RE
m6	10	48.51	69.46	Asymmetric 2008 shock per species + RE (Best Fit)

Conclusion: m6 is overwhelmingly the superior model. Capturing the unique, asymmetric impact of the 2008 shock across different species explains a massive amount of variance that standard random effects missed.

Results: Random Effects Model of Fishery Production

A linear mixed-effects model (LMM) was fitted using Restricted Maximum Likelihood (REML) to evaluate the impact of land use on fishery production ($\log(\text{prod})$) across five distinct species (nombre_principal), while controlling for species-specific baseline differences and an asymmetric 2008 macro-economic/environmental shock. Degrees of freedom and subsequent p-values were calculated using Satterthwaite's approximation method.

1. Random Effects and Variance Decomposition

The model successfully partitioned the random variance into species-level baseline variations and residual error, completely resolving previous model singularities.

- **Species-Level Heterogeneity (nombre_principal):** The random intercept variance across species is \$0.1871\$ (\$\text{SD} = 0.4325\$), indicating substantial baseline differences in production levels across the five species.
- **Residual Variance:** The remaining unexplained within-species variance is \$0.0735\$ (\$\text{SD} = 0.2711\$).
- **Intraclass Correlation Coefficient (ICC):**

$$\text{ICC} = \frac{0.1871}{0.1871 + 0.0735} \approx 0.718$$

Approximately 71.8% of the total unexplained variance in fishery production is attributable to intrinsic differences between species types rather than random annual fluctuations.

2. Fixed Effects and Environmental Drivers

The fixed-effect estimates revealed that both land-use types are highly statistically significant predictors of fishery production. Because production and land-use variables are log-transformed, the coefficients represent elasticities.

- **Natural Land (log_natural):** Exhibited a significant, positive relationship with production (\$\beta = 0.8400\$, \$t(48.00) = 3.916\$, \$p < 0.001\$). A 1% increase in natural land cover is associated with an approximate 0.84% increase in fishery production, signaling a strong ecosystem-service dependency.
- **Agricultural Land (log_agri):** Exhibited a highly significant, severe negative relationship with production (\$\beta = -2.4648\$, \$t(48.00) = -4.824\$, \$p < 0.001\$). Controlling for other variables, a 1% expansion of agricultural land correlates with an approximate 2.46% decrease in fishery production, suggesting severe negative externalities (e.g., runoff, habitat degradation).

3. Asymmetric Species Responses to the 2008 Macro-Shock

The interaction terms (shock_2008:nombre_principal) reveal that the macro-shock of 2008 did not impact fisheries uniformly, demonstrating high species specificity:

Species Cluster (nombre_principal)	Shock Estimate (\$\beta\$)	Std. Error	t-value	p-value	Significance	Impact Direction
Langostino	-3.7118	0.2841	-13.063	$< 2 \times 10^{-16}$	***	Catastrophic decline

Species Cluster (nombre_principal)	Shock Estimate (β)	Std. Error	t-value	p-value	Significance	Impact Direction
Mojarra	-1.7559	0.2841	-6.180	$\$1.31 \times 10^{-7}$	***	Severe decline
Jaiba	+1.6933	0.2841	5.959	$\$2.85 \times 10^{-7}$	***	Strong positive surge
Camaron	-0.2203	0.2841	-0.775	\$0.442	NS	No detectable impact
Robalo	-0.1560	0.2841	-0.549	\$0.586	NS	No detectable impact

- **Negatively Impacted Species:** Langostino experienced the most catastrophic reduction in production during the 2008 shock ($\sim 371\%$ decrease in the log scale), followed by a highly significant negative impact on Mojarra ($\sim 175\%$ decrease).
- **Positively Impacted Species:** Conversely, Jaiba (crab) experienced a highly significant, counter-cyclical *increase* in production ($\sim 169\%$ increase), hinting at ecological niche displacement or shifted market targeting during the shock year.
- **Resilient Species:** Camaron and Robalo displayed minor negative deviations that failed to reach statistical significance, indicating resilience to the 2008 macro-shock.

Summary Verdict: Model m6 confirms that preserving natural land cover and mitigating agricultural expansion are critical for protecting overall fishery yields. Furthermore, it demonstrates that broad economic or environmental shocks ripple through marine ecosystems asymmetrically, devastating specific crustacean and fish populations while leaving others unbothered or uniquely advantaged.